Lecture 20: The Good Life: Happiness (full transcript)

Source: https://oyc.yale.edu/psychology/psyc-110/lecture-20
License: Creative Commons Attribution-NonCommercial-ShareAlike 3.0 (Open Yale Courses)

Overview: The last lecture in the course wraps up the discussion of clinical psychology with a discussion of treatment efficacy. Does therapy actually work? Professor Bloom summarizes the different types of influences that clinical interventions might have on people who receive therapy. Professor Bloom ends with a review of one of the most interesting research topics in "positive psychology," happiness. What makes us happy? How does happiness vary across person and culture? What is happiness for? Students will hear how the most recent research in psychology attempts to answer these questions and learn how people are surprisingly bad at predicting what will make them happiest.

Chapter 1. How and Why Therapy Works [00:00:00]

Professor Paul Bloom: Welcome to the last class. Before starting the lecture I just want to do some final procedural issues. Again, your reading responses have been completed. Your essays have been all completed and the grades will be available sometime next week. The teaching fellows are currently grading them like crazy. Your experimental participation requirement is up until May 7th. Please finish it by then. If not — just please finish it by then. Your exam's on Monday. Last class — somebody asked me whether I was going to post an old exam and I said, "Well, you don't need it because you — I already posted a midterm and it looks like the Midterm," and one of the teaching fellows e-mailed me and essentially said, "Why are you so damn lazy?" [laughter] So, I posted the old final.

Some of you require special accommodations for the Final. The arrangements will be precisely the same as for the Midterm and Judy York should be contacting you to set up the arrangements. If there are any problems with that, definitely contact me as soon as possible but you should be hearing from Judy York concerning special arrangements.

Finally, the review sessions are going to be held tomorrow and on Friday. Please note: It turns out the original class schedule for the Thursday review session is booked. The review session will be held in this room from 6 o'clock to 8 o'clock and once again, the teaching fellows have graciously agreed to run these review sessions.

Okay. This lecture will be a some slightly shorter lecture than usual. What I first want to do is finish off the discussion of clinical psychology from last lecture and then have a little brief discussion about some very interesting research on happiness. We talked — we ended last lecture with a discussion of some early — some of the history of treating mental illness and we saw that it was rather gruesome, unsuccessful, and arbitrary. For the most part, we do better now, and Dr. Nolen-Hoeksema reviewed some of the therapies with focus on therapies for depression. The textbook talks in detail about therapies for different disorders including schizophrenia, anxiety disorders, and so on. The question which everyone is interested in is, "Does therapy work?" And this proves to be surprisingly difficult to tell. Part of the problem is if you ask people who go into therapy, "Did you get better after therapy?" for the most part they'll tell you that they did but the problem is this could be a statistical byproduct of what's called "regression to the mean."

So, the idea looks like this. This line plots how you feel from great through okay to awful and it goes up and down and in fact in everyday life you're going to — some days are going to be average, some days will be better than average, some days worse than average. You could plot your semester. You could do a plot every morning when you wake up or every night before you go to bed. You could put yourself on a graph and it'll come out to some sort of wiggly thing. Statistically, if something is above average or below average it's going to trend towards average just because that's a statistical inevitability. When do people go to therapy? Well, they go to therapy when they're feeling really crappy. They go to therapy when they're feeling unusually bad. Even if therapy then has no effect at all, if it's true that people's moods tend to go up and down after you feel really bad you'll probably improve rather than get worse. And so this could happen — the normal flow could happen just even if therapy has no effect at all.

And so, simply getting better after therapy doesn't tell you anything. On the worst day of your life you could do naked jumping jacks on the roof of your college for ten minutes. I guarantee you your next day would probably be better. That doesn't mean naked jumping jacks are helping you. Rather, it just means that the day after the worst day of your life usually is not as bad as the worst day of your life. It can get worse, but usually it just trends to average. What you've got to do then is you have to take people at the same point who would get treatment and compare them to people who do not get treatment or what we call a "control group." And this is an example of this. So, this is for people who are depressed. This is statistically equal. They start off pre-therapy. They all go for therapy but because in this example there's a limited number of therapists, some of them are put on a waiting list and others get a therapist. It's arbitrary. It's random, which is — which — making it a very good experiment. And in this example, you could see those who received cognitive training were better off. They had lower depression scores than those that received no therapy at all.

In general, in fact, we could make some general conclusions about therapy. Therapy by and large works. People in treatment do better than those who are not in treatment and that's not merely because they choose to go into treatment. Rather, it's people who are in desperate straits who seek out help. Those who get help are likely to be better off than those that don't get help. Therapy for the most part works. We can't cure a lot of things but we can often make them better.

Different sorts of therapy works best for different problems, and again, depression proves to be an illustrative example. If you have everyday unipolar depression, that is, you feel very sad and you show other symptoms associated with depression, an excellent treatment for you is some combination of cognitive behavioral therapy and possibly antidepressant medications like SSRIs. If you have bipolar depression, the cognitive behavioral therapy is useless but medication is your best bet and so on for all of the other disorders. Each disorder has some sort of optimal mode of treatment. If you suffer from an anxiety disorder, cognitive behavioral therapy can be of help. If you're a schizophrenic it's probably not going to be of much help at all. And so, different disorders go best with different sorts of therapies. Finally, some therapists do better than others. So, for reasons that nobody fully understands, there are good therapists and then there are better therapists and there are bad therapists. And there's great individual differences in the efficacy of an individual therapist.

Finally, putting aside then the difference in therapies and the difference in therapists, does it make sense to say that therapy, in general, works? And the answer is "yes." And this is in large part because

of what clinical psychologists describe as "nonspecific factors." And what this just is a term meaning properties that all therapies, or virtually all therapies, share and I've listed two of them here.

One of them is "support." No matter what sort of therapy you're getting involved in, be it a psychoanalyst or a behavior therapist or a cognitive therapist or a psychiatrist who prescribes you medication or someone who makes you go through different exercises or keeps a journal, you have some sense of support, some acceptance, empathy, encouragement, guidance. You have a human touch. You have somebody who for at least some of the day really cares about you and wants you to be better and that could make a huge difference.

Also you have hope. Typically, there's an enthusiasm behind therapy. There's a sense that this might really make me get better and that hope could be powerful. Sometimes this is viewed under the rubric of a placebo effect, which is that maybe the benefits you get from therapy aren't due to anything in particular the therapist does to you but rather to the belief that things are going to get better, something is being done that will help you. And this belief can be a self-fulfilling prophecy. "Placebo effect" is often used sort of in a dismissive way, "Oh, it's just a placebo," but placebos can be powerful and even if it's useless from a real point — from a psychological theory point of view, even if the therapist runs around and dances while you – I have dancing on my mind now – while you sit in the chair and watch him dance; if you believe the dancing is going to make you better, it may well help.

Chapter 2. Question and Answer on Therapy [00:07:48]

Okay. That's all I'm going to say about therapy. Any questions about therapy? Yes.

Student: [inaudible]

Professor Paul Bloom: Fair enough. The question is the assumption of regression to the mean seems sort of arbitrary because it depends what the mean is. Always after the fact you can apply an average to it and say, "Look. This is the average," but how do you know beforehand? It's a good point. When you talk about regression to the mean, it adopts certain assumptions. The assumption is there really is an average throughout much of your life and things go up and down within that average and for the most part that's true for things like mood. For most of us, we have an average mood and we have bad days and we have good days. It's always possible that you have a bad day and then from there on in it's just going to go down and down and down but statistically the best bet is if you have a bad day you're going to go back up to the mean. It's — in some way you don't even have to see it from a clinical point of view. You could map it out yourself. Map out your moods and the days where you're most depressed sooner or later you're likely to go up. Similarly, on the happiest day of your life odds are the next day you're going to go down and there's nothing magical about this. This is just because under the assumption that there really is an average in — built into one — each of us. If human behavior was arbitrary, it would be like a random walk but it's not. We seem to have sort of set points and aspects of us that we fall back to that make the idea of a mean a psychologically plausible claim. Yes.

Student: [inaudible]

Professor Paul Bloom: That's a good question. Yes. In that study it's a perfectly good hypothesis that the sort of anxiety of being told, "I see you've come here for help. We can't give it to you. Congratulations. You're a control group" [laughs] causes anxiety. In other studies the control group doesn't know they

are the control group. So sometimes you can do an intervention where you say, "Congratulations, everybody in Intro Psych who did very low on the depression inventory," which many of you filled out, "We're going to do something to you." And then the rest of the people don't even know that they haven't been chosen. So, you're right. It's a perfectly good point. Knowing you're not chosen could have a deleterious effect and the way to respond to that is you have other studies that don't use that same method.

Chapter 3. Happiness and Positive Psychology [00:11:16]

Okay. I want to end with happiness and it's a strange thing to talk about in psychology. Most of psychology focuses on human misery, most of clinical psychology. There is the psychology we spoke about through most of the semester on vision and language and social behavior, but typically when people think about interventions what they think about is people having problems and then we figure out how to make them better. They are schizophrenic, they are depressed or anxious, they are just not making it in life, and psychologists try to figure out how to make things improve. And in fact, a lot of the information I gave you at the beginning of the lecture last class where I reviewed all of the disorders is in this wonderful book called DSM-IV, The Diagnostic and Statistical Manual of Mental Disorders. If you ever really want to get — If you really [laughs] want to diagnose people and come to have a belief in your own mental instability, browsing through that book is a treat. Everything that can go wrong in mental life from Aspergers syndrome to fetishes to paranoid schizophrenia is all in that wonderful book and — but a lot of psychologists have been disturbed by the focus of our field on taking bad people, people who are broken, people who are sad, and bringing them up to normal. And they've started to ask can psychology give us any insight into human flourishing, how to take people who are — who — how to study people who are psychological successes, how to take people who are psychologically okay and make them better. And this is the movement known as "positive psychology." And it has its own handbook now, The Handbook of Positive Psychology, listing psychological strengths, listing virtues, ways — what psychology tells us about how we can be at our best.

Some of this work in positive psychology is, in my mind, real crap. A lot of it is some combination of new age banalities by people who are striving to get more grant funds and end up on Time magazine. On the other hand — and so, some of it is really bad. You could imagine this attracts every self-help huckster you could imagine. On the other hand, a lot of this work is quite neat, quite interesting and quite promising. And what I want to do is tell you what I think is the most interesting research from this movement concerning happiness.

Now, there are a lot of good books on this and I'm going to recommend books, which I haven't been doing much in this class. Marty Seligman is the pioneer of positive psychology and he's written an excellent book called Authentic Happiness. Jonathan Haidt is a brilliant young scholar who's done — also done a lot of work on disgust and morality. He did the "sex with dead chicken study" we discussed earlier. This is one of my favorite books by – Happiness by Nettle, because it's smart, it's beautifully written and it's extremely short. And Dan Gilbert's book, Stumbling on Happiness, is a very, very funny book and very smart book and is now on The New York Times bestseller list. So, there's no shortage of books on happiness.

So, the starting point is — And a lot of research on happiness starts with a basic question: How happy are you? And we're psychologists so tell us on a scale of one to ten where five is average, ten is super-duper. The most common answers, interestingly enough, are high. They're seven or eight. How many people in this room would give themselves a seven or an eight? Okay. How many a nine or a ten? All right. How many a ten? Good, good, maxed out on happiness. It turns out that most people think that they're pretty happy. There's a Lake Wobegon effect with happiness. Most people think they're very happy. In fact, most people think they're happier than most people, which shouldn't really happen.

This question, "How happy are you from one to ten?" has been asked all over the world. So — and it turns out there are slight differences depending on how old you are. There are slight differences depending on your place within a country, California versus New York. There are slight, subtle differences between men and women at different points, somewhat paradoxically. Although women are more vulnerable to depression than men, still on average women are slightly happier than men. The country-by-country data is quite interesting. In one study they looked at forty-two countries. The happiest — well, let me see. The happiest people on earth — well, first, no country believed they were unhappy, the people in no country of these forty-two countries. I mean, you're thinking there are some really bad countries to live in and I don't know if they were tested but of these forty-two everybody seemed — said they were above average. The happiest people on earth? The Swiss. [laughter] They think — they're just like — they're just so happy. I was talking to people about this last night and they suggested chocolate. [laughter] The saddest people on this — on the sample? The sad Bulgarians. [laughter] You are wondering what about Americans. Americans are actually pretty happy, 7.71. We are a happy country full of happy people.

Now, I'm going to talk about a lot of research that's based on the data you get when you ask people how happy they are from a scale of one to ten. But I'm going to be honest and tell you there are reasons to be cautious about these numbers. And the reasons come from a couple of experiments. In one experiment they asked people inside a psychology department where there was a photocopy machine. They went up to people — the people were going up to the photocopy machine to make copies and when they were done making copies they asked them, "How happy are you with your entire life?" There were two groups. Group A, they put a dime on top of the photocopy machine so people walked over there, "I'm going to [inaudible]. Oh, a dime. Well." The other group, no dime. It turned out that when asked "How good is your whole life?" [laughter] group A reported [laughter] greater life satisfaction overall in their entire lives.

Another study asked people how happy you are with your whole life on sunny days like today and people said they were happier on sunny days than rainy days. What's interesting is you could make this effect go away if you ask immediately before "How's the weather?" These were done by phone interviews. And logically, what seems to go on is that if you're asked how's the weather, you're "Oh, it's really sunny outside," and then when people are asked "How happy are you with your whole life?" people then say, "Oh, okay. I'm going to take into account the sunny-ness when I give my answer.

Chapter 4. Getting Used to Happiness [00:18:59]

Okay. So, what is happiness? What are people rating when they're answering these sort of questions? And this is an extraordinarily difficult question and one could devote a seminar to discussing it, but

one simple answer from an evolutionary point of view is that happiness — forget about "what is happiness?" Ask "what's happiness for?" just like we've asked what language is for, or what laughter is for, or what hunger or lust is for. What's happiness for? And one answer is happiness is a goal state that we've evolved to pursue. It's a signal that our needs have been satisfied. Happiness is the carrot we're running towards that makes us take care of our lives. We want to be happy. An example of this is food. You're not very happy if you're starving. You want to be satiated, you want to be satisfied, so you seek out food to fill your belly. Once you've done it, you're happy.

Steven Pinker summarizes the keys to happiness in a nicely evocative passage: "We are happier when we are healthy, well-fed, comfortable, safe, prosperous, knowledgeable, respected, non-celibate, in love." How many people here have got all of those right now? Oh, come on. [laughter] Some people. Oddly enough, the person who said he was a ten didn't — does not raise his hand. Okay. [laughter] So this is- And this makes out — you get all your needs satisfied, your belly is full, people love you, you're getting sex regularly, you're smart, you're rich, you're happy, but as Pinker points out it's not that simple.

Here's the problem. You, Americans in this century, you are now healthier, better fed and so on than just about anyone in history but you're not happier. That's the puzzle. In particular, these studies asking about happiness have been around for a long time. People in the 1950s did not make as much money, did not eat as well, did not live as long, suffered from more diseases, were more vulnerable in a hundred different ways, yet they were — are as happy as you are now. You are as happy as your parents were and they were as happy as your grandparents. Moreover, in poor countries people don't have the shelter, the knowledge, the protection, the safety, yet, for the most part, there's not a huge effect on how rich a country is and — on how happy the people are.

Furthermore, there are great individual differences in happiness among people whose basic needs are met. For the most part, everybody in this room is fed and sheltered and safe. Some of you are prosperous, some of you are knowledgeable, a couple non-celibate, and — [laughter] but even among that group you vary in your happiness, and that's kind of a puzzle. And to explain the puzzle we need to talk about a few surprising facts about happiness and I'll present three of them.

The first is happiness doesn't change as much as you think. In particular, happiness is not as sensitive to what happens — your happiness is not as sensitive to what happens in your environment as you might think it is. Part of the reason for this is that there appears to be a strong heritable basis for happiness. So, just as we talked about the domains of personality and intelligence, there is some genetic determination, not entirely but some, in how happy you are. And some people talk in terms of a genetically determined set point. So, you have a sort of natural happiness level, maybe a range. To put it in extreme form, some people are genetically predisposed to be pretty sour, others to be pretty cheerful. Well, that can't be it. Identical twins are very similar in their happiness but, as with everything else we've discussed, they're not identical. What about life events? Wouldn't life events change your happiness? And here we're entering one of the great discoveries of happiness research.

Think for a moment. What's the worst thing that could happen to you? And then ask how much would it change your happiness. Now, think for a moment. What's the best thing that could happen to you? And ask how much would it change your happiness. And the research in happiness suggests that your gut feelings are probably wrong. And here's a couple of case studies.

For many people a very bad thing that could happen to you is to be paralyzed from the neck down in an accident. It turns out obviously, common sense, that when this happens it makes people very unhappy. It makes them depressed, they think their life is over, they feel terribly sad, but not for that long. After about a year after being paralyzed from the neck down, people's happiness comes back up pretty much to where it was before, suggesting that there's a temporary effect but not a permanent one. Many people believe that winning many, many millions of dollars in the lottery will make you happier and it does. When you open up that winning ticket and you say, "I won one hundred million dollars," you say, "Woo, hoo!" You are honest to God very happy. You'd say, "Hell, I'm a 10.5, I am very happy." A year later you are not as happy. In fact, lottery winning may be a terrible case where people — where it goes the reverse of what you expect. What happens when you win a lot of money is it often wrenches you away from your family, your work and your friends and leads you to depression and sadness but even mundane events that would make you happy — that you think would make you happy don't seem to last.

In some research by Dan Gilbert and others, they've asked young assistant professors who are coming up for tenure, and tenure in a university system is a good thing to get because it gives you lifetime job security, "How happy would you feel if you got tenure? How happy would you feel if you didn't get tenure?" Prior to the last election, they asked people "How happy would you be if it was President Bush? How happy would you be if it was President Kerry?" And it turns out people radically overestimated the effects of these things. Having your favorite candidate win is not such a big deal. Having your favorite candidate lose is not such a big deal either. Getting tenure or not getting tenure are really big when it happens. Six months later and a year later your happiness doesn't seem to be affected.

The purchase of consumer goods, an Xbox 360, a nice flat screen TV, those sorts of things make you very happy when you open up the package and set it up but this happiness fades almost immediately. The moral of — A lot of people are shaking their heads. It's true, not for me but — the moral of a lot of this work is we think these things will have big permanent and profound effects but they need not and they often don't.

Why not? Why do we overestimate their happiness? And the technical term for this, by the way, is "affective forecasting." Again, this is Dan Gilbert's work and the idea is we are bad at affective forecasting. That is, we are bad at predicting how happy or sad we will be in the future based on what's happening to us. Why? Well, a couple of reasons. One thing is there's often a failure to appreciate the day-to-day irrelevance of certain events. So prior to the election — the election's happening tomorrow and I ask somebody who's a diehard Democrat, "How would you feel if Bush won?" and the person said, "I'll be miserable. It'll be a miserable four years afterwards." But what often isn't appreciated here is that whether or not Bush wins will make you sad or happy after it happens but for the most of your day-to-day life you aren't thinking about who the president is. I'd be very happy if I won a huge prize and I'd be "Whoa, a huge prize," the Nobel prize, a Guggenheim, a MacArthur ‘Genius' or I get them all in the same day. [laughter] "What a day. I am really happy." But then a month later I'm there and I've still got my regular insomnia and there's nothing on TV and the plumber's not coming and my kids don't respect me and I can't — and the fact that "Yeah, but I won the prize," it doesn't matter. A lot of the things in life that'll make you really — that you think will make you really happy don't have this day-to-day effect.

Also, there is the logic of the set point. And this comes to a terrible word: We adapt. Right now I'm a guy without a Nobel prize. I'm kind of used to it. If I got a Nobel prize I'd be a guy with a Nobel prize. I'd be happy but then I'd kind of get used to that too. And if I got a second one, "whoa, two!" but then I'd just get used to that too. You get used to things. You get used to bad things.

Now, I don't want to overstate this. There are some very interesting exceptions. So for instance, we don't get used to noise. A lot of research suggests that if your environment is noisy, they're doing construction around you, you can't get used to it. Your happiness drops and it doesn't come back up. Your system cannot habituate to continued noise. We adapt to good things, winning the lottery, winning a prize, getting an "A " in a course. We adapt, we get used to it, also with some surprising exceptions. One of the big — one of the other surprises from happiness research is the effects of cosmetic surgery like breast enhancement and breast reduction. One of the big surprises is it makes people happier and then they stay happier. And one explanation for this is how we look is very important. It's very important for how other people see us and how we see ourselves, and you never get used to looking in a certain way. So, if you look better it just makes you happier all the time.

So, there are these exceptions but putting aside the exceptions, the problem of adaptation is sometimes called "the hedonic treadmill" and the idea is hedonic for happy. You keep on running but no matter how fast you run you stay where you are, you get used to it. Habituation is like you put — you step into a very hot bath but you get used to it. If it's a cold bath you get used to it. A difficult environment, an easy environment, you get used to it. The story is often illustrated — It's often illustrated with a story from the Bible in Ecclesiastes of a king and this king had it all. He had gardens, parks, vineyards, castles, slaves and concubines and they were both male and female concubines. [laughter] So, he had everything, right, but it didn't make him happy and here's what he says: "I hated life. All this vanity and a chasing after wind and there is nothing to be gained under the sun."

Now, in these books I talked about at the beginning these authors give advice on how to deal with the hedonic treadmill. How do you deal with the fact that everything you aspire to, once you get it you'll be used to it and it will lose its value? Well, one answer is that possessions are not the key to happiness. Possessions you very quickly get used to. From there, there are two alternatives. One is endless novelty. So one guy – I forget his name – wrote a book and he says, "Look. There's the hedonic treadmill. The trick is always do something different. Next week have sex with somebody you've never had sex before. Then climb Mount Everest. You get bored with that, become an accountant. Boring. Scuba diving. Boring." He had endless ideas, and that's a possibility. You never — you — if you keep changing what you're doing you'll never get used to anything and you'll always be happy. At least he says that. Then there's the old guy alternative. Step off the treadmill. Give up — give up chasing the whole happiness thing and then seek out more substantial goods that might actually not make you happy in the simple sense of a quick fix of delight, but substantial goods like friends and family and long-term projects.

So, the first moral of the science of happiness is that your happiness is actually rather fixed. It's fixed in part genetically and it's fixed in part because what happens in your life you'll get used to, to a large extent. Are you raising your hand? No. Oh, sorry. The second one is happiness is relative. So, there's a lot of research on money, power and happiness, and remember I did say before that it doesn't matter whether you come from a rich country or a poor country. As long as your country — as long as you're not starving to death, it kind of doesn't matter how rich your country is for how happy you are. But

that's not the same as saying it doesn't matter how much money you make. In fact, there's a set point or a range but there is some effect on your salary and on your job on your happiness. And if you're desperately poor, no matter where you are, no matter who's around you, you're not going to be happy. But beyond that your happiness depends on your relative circumstance. And this is an old insight. H.L. Mencken wrote, "A wealthy man is one who earns a hundred dollars more than his wife's sister's husband." The idea is what matters isn't how much you make. What matters is how much you make relative to the people around you.

And they've asked people this. "What would you rather? Do you want to make seventy thousand dollars if everybody else in your office is making sixty-five thousand or seventy-five thousand dollars if everybody else is making eighty thousand?" Does it matter how much money you bring home or does it matter how much money you make relative to other people? Well, they're both factors but relative salary — and in this example people prefer this. [pointing to the relative choice] They prefer to be making less if they're making more than the people around them. It turns out that there's research on British social servants and their happiness and their health and the quality of their relationships and how they love their lives doesn't depend on how much money they make. It depends on where they are relative to everybody else.

We are very status conscious primates and your role in a hierarchy, your level in the hierarchy plays a — has a significant effect on your level of happiness. This is not really a secret. The opera star Maria Callas and the English professor Stanley Fish had the same negotiating strategy. When Fish got hired into his department, according to urban legend at least, he said, "I don't want to talk salary. I don't have a particular number in mind. I just want to get paid one hundred dollars more than whoever is the top person in this department." And that's a guy who knows about happiness. He walks in and he states, "I'm paid more than everybody else. I don't care how much it is. It's just more." [laughter] And that's relevant to happiness.

We're now in a position to give some advice to the king, summing up. First, going back again to Dr. Nolen-Hoeksema's lecture, I think the king is suffering from mild unipolar depression so we should — he should get some SSRIs and cognitive behavior therapy. I think he needs to move his castle to a quiet part of the kingdom. The noise of a busy castle is stressful. And he needs to give up on the concubines. He needs to find a queen. He needs to develop social relationships, join a club, get involved in charity, maybe a hobby. The final finding is a bit of a jump to a different topic but — sorry. You raised — yes.

Student: [inaudible]

Professor Paul Bloom: The question is, "Would you become used to changes of social interaction?" An example is solitary confinement or prison. People get used to a lot of things. It could be that in those examples there is a shift in status as well and social relationships and how you think of yourself relative to others so that may be difficult to recover from. If you're popular at Yale and you go to prison you're not going to be very happy your first few days in prison probably. But then suppose you get popular in prison, people like you, you're the head of the book club and everything. [laughter] You'd rather not be in prison — been in prison but you'll probably be a pretty — if you're a cheerful guy here you'll be a cheerful guy in prison.

The final case is — involves judgments of the pleasure and pain of past events. And I mentioned the Nobel prize a little while ago as an example but the work I'm going to talk about is actually from the one psychologist who's alive who's won the Nobel prize — sorry, one of the ones who's won the Nobel prize, Daniel Kahneman. And he became interested in happiness. And you remember him from his rationality research and this is some work on happiness. Here's an example. Anybody see the movie Marathon Man? It involves a dental torture scene. Imagine you're going to the dentist and he's torturing you or it's a cleaning but it's really painful and it's terrible. You're sweaty, you're squeezing the thing, and it lasts for an hour and then when it's done the dentist leans over and says, "We're done now. If you want we could stop. But if you want, as a favor to you – I have five minutes – I could top you off with some mild pain." This may seem like a very odd thing to ask. Here's the alternative. What do you want? A dental procedure that's very painful for an hour or a dental procedure that's very painful for an hour — sorry about the "S" there [referring to a slide typo] — and then some additional mild pain? Who votes for A? You got to choose one. Who votes for A? Okay. Who votes for B? Okay.

Here's the big finding. The big finding is "yes," B involves more pain. It's true. "A" is an hour of pain. "B" is an hour of pain plus a few minutes of more pain. Seems like a no-brainer but when you have A or B and you have to remember it later on you'll have a much nicer memory of B than A. Kahneman's insight is when you think back on events you don't just add up the amount of pleasure or pain you experienced. Rather, your memory is highly skewed to peaks and then to endings. And you could imagine this. In the first case, "A," you leave and say, "Oh, God, that was terrible. Oh." "B," you leave and say, "Oh, that was mildly painful. There was something terrible in the middle there but it ended okay, mild pain." It turns out, in general, endings matter a lot.

Kahneman did his research with – both in a laboratory where you could make – give people mild pain by getting them to stick their hand in freezing cold water and with people undergoing extremely painful colonoscopy procedures. And it turns out that if you want to give people a good memory, or a less bad memory, of a horrible event topping them off with some mild pain will do the trick. In the end, endings matter. Both of these examples, a party that's hugely fun for ninety percent, then the last ten percent somebody slaps you in the face and pours dip on you or something — [laughter] So, ninety percent of good stuff, ten percent bad stuff, versus ninety percent people are slapping you and pouring dip on you [laughter] but then ten percent, whoa, that was a really good — when you think back on it, if you just added it up, "A" would be much better but "B" has this huge pull because of the power of how things end so endings matter.

Chapter 5. Closing Remarks [00:42:41]

So, I'm going to end things now. I'm first going to do a few things. Before saying anything more, I want to thank the teaching fellows. There's Sunny Bang, Erik Cheries, Jane Erickson, Izzat Jarudi, Greg Laun, and Koleen McCrink. I think they did a superb job. [applause] We've basically reviewed all the psychology. Here is a promissory note we started with at the beginning of this semester. I think you are now in a position to answer or at least consider answers about these topics, about topics such as dreams, testimony, disgust, memory, depression, language, humor, and even a little bit about good and evil. This is an — a broad intro survey class, and the field of psychology is broad and we've just gotten started. If you're interested, this is a great department. There are some amazing scholars here and

some amazing teachers and there are courses that go into detail about just about every topic I talked about. If you're interested in memory or social interaction or mental illness, I could point you to some great courses. I'm not taking any sophomore advisees next year because I'm on leave in the fall but you should feel free to come talk to me if you want any specific advice or suggestions.

Now, I know not all of you are going to end up majoring in psychology. Some of you will choose cognitive science instead [laughter] but on a more serious note I know for some of you this is the last — maybe the first but the last psychology class you'll ever take. And so I want to close this course by emphasizing two themes. The first one is a bit of humility. There are some very basic questions about the mind – and I've tried to be honest about this throughout the course — There are some very basic questions about the mind that nobody knows the answer to yet. We know the brain is the source of mental life but we don't have any understanding at all about exactly how this happens, about how a physical object, a lump of meat, can give rise to conscious experience.

We know that about half of the variants in personality, about half the differences between people, are due to genetic factors but we don't know how to explain where the other half happens. It has to be experienced but we have no real good theories of the sort of experience that makes one person adventurous and another one timid, one bitter and one satisfied. We know a lot about the social influences that can drive people to do terrible things to one another but we don't know the answer to the maybe harder question of why some of us — some people are immune to these influences, why some people do good things, perhaps even heroic things, regardless of the circumstances that they find themselves in. So, there's an enormous amount left to do. It's an exciting field just because there's just so much more we need to understand.

The second theme is more optimistic. And this is the idea that we're going to eventually come to answer these questions and many more questions through the sorts of methods we've been discussing this semester, through constructing scientific theories, evolution — evolutionary, neurological, developmental, computational and testing them through experimental and observational methods. This is the idea that, in the end, the most important and intimate aspects of ourselves, our beliefs and emotions, our capacities to make decisions, even our sense of right and wrong can be explained through constructing and testing scientific hypotheses. Now, the reason why I'm optimistic is I think there's been some success stories where we really have learned some surprising and important things about the mind and there's no reason to expect this way of proceeding to fail us in the future.

In my very — in the very first class on the brain I ended by talking about people's worries here, and I'll be honest, that some people find it a scary prospect. Some people believe that a scientific approach to the mind takes the "special-ness" away from people, that it diminishes us somehow. And I don't agree. If there's anything I've tried to persuade you through the course, it's that the more you look at the mind and how it works from a serious scientific point of view the more you come to appreciate its complexities, uniqueness, and its beauty. This has been a great course to teach. Thank you for coming and good luck on the Final. [applause]